

EN.553.171 Discrete Math
Fall 2026 Syllabus

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1 Course Information

Course: EN.553.171 Discrete Math, Fall 2026

Instructor: Apurva Nakade, apurva.nakade@jhu.edu

Office hours, TA information, and course materials will be posted on Canvas.

1.1 Class Meetings

Section	Class Time	Location	Discussion Time	Location
Section 1	MWF 12:00–12:50 PM	Bloomberg 274	Th 12:00–12:50 PM	Gilman 17
Section 2	MWF 12:00–12:50 PM	Bloomberg 274	Th 1:30–2:20 PM	Bloomberg 274

Please check SIS for any updates to the schedule.

1.2 Discussion Sections

Discussion sections meet weekly on Thursdays and are an essential part of the course. In discussion you will review the week’s material and start working on the next homework in groups, with a TA on hand to help. **The main purpose of these sections is to build a community of students to work with and to learn from each other.** Use this time well. Getting started on the homework here, with your classmates and a TA in the room, is much easier than facing a blank page alone the night before it is due.

1.2.1 Class Expectations

The class meets three times per week for 50 minutes, mixing lecture and discussion. Participation is not graded, but we strongly encourage you to attend and take part anyway: it is genuinely the best way to learn the material, and it’s where you’ll meet the classmates you’ll work with all semester. Come prepared having done the reading, ask and answer questions, and join in the group work.

If lateness becomes a pattern, we’ll ask to meet with you to figure out what’s getting in the way.

Exception: if another class or work commitment prevents you from arriving on time, let us know well in advance.

1.2.2 Devices in Classroom

Laptops and phones are not allowed in class. You will be given a warning the first time and asked to leave the classroom the next time. If you have an approved accommodation from Student Disability Services for assistive technology, this restriction does not apply to you. Just let us know in advance so we’re aware.

Tablets are allowed for note-taking only. In fact, you are encouraged to take notes on a tablet, as proof writing is an iterative process requiring multiple revisions. Taking notes on a tablet will allow you to easily revise your proofs.

1.3 Course Websites

This course uses a few separate websites. Each one has a specific purpose:

Site	Used for	What you'll find there
Canvas	Announcements	Course announcements, office hour times and locations, the syllabus
Gradescope	Written homework and exams	Homework submission, graded feedback, and exam scores
zyBooks	Textbook	The textbook itself and all textbook assignments (see section 4)
OneNote	Lecture Notes	Updated in real time

These are also posted on Canvas. **A note on Canvas:** Canvas does not reliably sync assignment and exam scores and dates from the written-work platform or zyBooks, so we will not use Canvas to track assignments or grades. Instead, we will make regular weekly announcements about due dates.

1.4 Resources and Communication

- **Office Hours:** Times and location will be posted on Canvas
- **TA Office Hours:** Posted on Canvas
- **Email Policy:**
 1. Email the instructor listed above.
 2. Include “[Discrete Math]” in the subject line.
 3. Allow 24–48 hours for response (longer on weekends).

1.5 Course Description

Introduction to the mathematics of finite systems with emphasis on proof writing and mathematical reasoning. Topics include logic, direct proof, proof by cases, proof by contradiction, sets, relations, number theory, modular arithmetic, applications to cryptography, combinatorics and combinatorial proof, mathematical induction, and graph theory. The concept of a proof and development of the ability to recognize and construct proofs are critical parts of the course. See [section 2](#) for the detailed schedule, which is the authoritative list of topics for this offering.

Recommended Course Background: Four years of high school mathematics. *Fluency in high school algebra is a strong prerequisite for this course.*

1.5.1 Course Philosophy

You will notice that this course has a lot of assessments, running on several different clocks at once:

- **Textbook assignments**, due before most classes
- **Written homework**, almost every week
- **Midterms**, roughly every month
- **A cumulative final exam**, once at the end

This is deliberate. The course is built around **active learning** and **spaced repetition**: you learn mathematics by doing it yourself, and you retain it by returning to the same ideas over weeks rather than meeting them once. Each idea will come back at you repeatedly, first in the reading, then in class, then on the homework, then on an exam.

Think of it like the gym, or learning the piano. Nobody gets stronger by reading about lifting, and nobody learns a piece by hearing it played once. You build the ability by practicing badly at first, then less badly, again and again, with rest in between. The proof techniques you’ll learn have to become close to muscle memory before you can reach for the right one on an unfamiliar problem.

The practical consequence: **this course cannot be crammed**. Do a little every day and the workload is manageable. Save it for the night before and it will not be.

1.6 Learning Objectives

There are two main learning goals in this course:

1. **Learning to write proofs.** Mathematics is built on rigorous logical reasoning, not just calculation. You'll learn to build clear arguments and express mathematical ideas precisely, skills that will serve you in every advanced mathematics and computer science course you take from here on. Nobody arrives already knowing how to do this, and by the end of the semester you'll be surprised how much your writing has sharpened.
2. **Creative problem solving.** You'll learn to approach unfamiliar problems without being told in advance which method to use, building a genuine toolkit of techniques and the confidence to choose from it, adapt when a strategy stalls, and craft solutions that are truly your own.

This course is your entryway to advanced mathematics: you'll find that mathematics is not algorithmic or mechanical, and you'll pick up the shared language of proof that mathematicians everywhere use. It is a 100-level course, because it is aimed at *first and second year* students. One of the most valuable things you'll take away is the connections and friendships you make with other students here, many of whom you'll keep learning alongside throughout your undergraduate journey.

1.7 Strategies for Success

In terms of difficulty, this course is at least a 200 level course. Learning to write proofs and solve mathematical problems takes time and practice, and everyone gets there at their own pace. Please don't hesitate to seek help when you need it! Asking questions is a sign of engagement, not weakness. Some of the best mathematicians are the ones who are most comfortable saying "I don't understand this yet" and asking for clarification.

Here are some strategies that past students have found genuinely helpful:

- **Start homework early** – Proofs reward multiple attempts and time to think; starting early gives your ideas room to develop
- **Practice regularly** – Work through textbook examples often, covering the solutions and recreating the proofs yourself
- **Form study groups** – Explaining a proof to someone else is one of the fastest ways to deepen your own understanding
- **Focus on reasoning** – Aim for the "why" behind each step, not just the format; the format will follow
- **Keep a techniques reference** – Collect proof methods (direct, contradiction, induction) as you learn them and watch your toolkit grow
- **Use office hours** – You don't need to be stuck to come by! Bring specific questions, or just come to work through problems. We'd love to see you there.
- **Embrace the struggle** – Finding proofs hard at first is completely normal and a sign you're learning something real; take a break when frustrated and come back fresh
- **Use AI for learning and not cheating** – See [section 6](#) and [section 7](#).

Remember: The goal is steady improvement in your mathematical reasoning, not immediate perfection. Every mathematician you admire started exactly where you are now.

2 Course Schedule

The schedule below is tentative and subject to change.

2.1 Weekly Schedule

Week	Dates	Topics	Notes
1	Aug 31 – Sep 4	Introduction; Logic; Quantifiers	HW 0 due Thu
2	Sep 7 – 11	Direct Proofs; Inequalities; Proof by Cases	No class Mon; HW 1 due Thu
3	Sep 14 – 18	Set Theory	HW 2 due Thu
4	Sep 21 – 25	Relations	HW 3 due Thu
5	Sep 28 – Oct 2	Number Theory; Modular Arithmetic	HW 4 due Thu
6	Oct 5 – 9	Multiplicative Inverses	Exam 1 (Mon)
7	Oct 12 – 16	Diffie–Hellman Key Exchange; RSA	HW 5 due Thu
8	Oct 19 – 23	Proof by Contradiction	Fall Break Thu–Fri (no class) HW 6 due Wed
9	Oct 26 – 30	Counting	HW 7 due Thu
10	Nov 2 – 6	Stars and Bars	Exam 2 (Wed)
11	Nov 9 – 13	Combinatorial Proofs; Mathematical Induction	HW 8 due Thu
12	Nov 16 – 20	Introduction to Graphs; Trees	HW 9 due Thu
–	Nov 23 – 27	Fall Recess	No class
13	Nov 30 – Dec 4	Euler Circuits and Hamiltonian Cycles; Planarity; Six Color Theorem	HW 10 due Thu
14	Dec 7 – 11	Euler Characteristic	Exam 3 (Fri); HW 11 due Fri

2.2 Homework Topics

Homework	Topic	Week	Due Date
HW 0	Syllabus Quiz	1	Thu, Sep 3
HW 1	Logic	2	Thu, Sep 10
HW 2	Introduction to Proofs	3	Thu, Sep 17
HW 3	Proof by Cases, Inequalities	4	Thu, Sep 24
HW 4	Sets, Relations	5	Thu, Oct 1
HW 5	Number Theory	7	Thu, Oct 15
HW 6	Cryptography	8	Wed, Oct 21
HW 7	Proof by Contradiction, Counting	9	Thu, Oct 29
HW 8	Combinatorial Proofs	11	Thu, Nov 12
HW 9	Induction	12	Thu, Nov 19
HW 10	Graph Theory	13	Thu, Dec 3
HW 11	Graph Theory	14	Wed, Dec 9

Note: “Cryptography” is used throughout this syllabus as an umbrella term for the Diffie–Hellman key exchange and RSA.

3 Grading

The following is the breakdown of the grading scheme:

Assessment Type	Component	Percentage
Formative Assessments	Textbook Assignments	10%
	Written Homework	20%
Summative Assessments	Midterms (best two of three, 17% each)	34%
	Midterm (lowest of three)	11%
	Final Exam	25%

At the end of the semester, grades will be assigned such that the median grade is *at least* B. Grade assignments will generally follow university standards but are ultimately at our discretion. Final grades are not based solely on arbitrary percentage cutoffs but also reflect your overall learning, improvement over the semester, and engagement with course material.

3.1 Textbook Assignments

Graded assignments are embedded in the online textbook and are due before each class. See [section 4](#) for how to access the textbook and [subsection 4.1](#) for the late-submission policy.

3.2 Written Homework

Written homework assignments will be posted weekly and must be submitted via Gradescope. Late homework will not be accepted for any reason. Detailed submission instructions and formatting requirements will be provided on Canvas.

3.2.1 Resubmission Policy

Learning math is an iterative process. You are not expected to get everything right on the first attempt, and mistakes are a natural part of learning. To encourage this, you may **redo homework problems to recover all lost points**. Bring your graded work to TA office hours within two weeks of the deadline, discuss what went wrong, and rewrite the solution from scratch with the TA.

This policy only applies to good faith attempts. Which means that you should submit whatever work you have, even if it is incomplete. Include the definitions you believe are relevant, explain your approach, and note where you got stuck. A partial attempt still gives us something to comment on and keeps you eligible for a redo later. **A missing submission, a blank page, or work that is not genuinely your own (including copied solutions or AI-generated work) does not qualify for resubmission.**

3.2.2 Collaboration Policy

You are encouraged to discuss homework problems with classmates, but all submitted work must be your own. You must write your own solutions and proofs. Copying solutions from any source is prohibited. If you worked with a few friends on a homework assignment, please report it. This helps us distinguish legitimate collaboration from copying. You are very welcome to discuss problems with your friends, but you should not copy anyone's work.

3.3 Exams

The course will consist of three midterm exams and a final exam. The dates and times are as follows:

Exam	Date	Time & Location
Midterm 1	Monday, October 5	In class
Midterm 2	Wednesday, November 4	In class
Midterm 3	Friday, December 11	In class
Final	Monday, December 21	6:00–9:00 PM, location [TBA]

Midterm exams will be held during class time and will not be cumulative. The final exam will be cumulative and will be held during the final exam period. If illness or a personal emergency causes you to miss an exam, see [subsubsection 3.3.3](#).

The date and time of the final exam are set by the registrar, not by us. Please plan any end-of-semester travel around it. Travel plans are not grounds for taking the final exam early or late.

Exam accommodations. Students with exam accommodations through Student Disability Services must schedule and take their exams with SDS. Please arrange this with SDS well in advance of each exam date.

3.3.1 Exam Topics

Exam	Topics Covered	Homework Covered
Midterm 1	Logic, Direct Proofs, Proof by Cases, Inequalities, Set Theory, Relations	HW 1–4
Midterm 2	Number Theory, Modular Arithmetic, Multiplicative Inverses, Cryptography, Proof by Contradiction, Counting	HW 5–7
Midterm 3	Advanced Counting (Stars and Bars), Combinatorial Proofs, Induction Proofs, Graph Theory	HW 8–11

3.3.2 Exam Conflicts

If any of the exam times conflict with your schedule, you must let us know as soon as possible. The strict deadline for this notification is **three weeks prior to each exam date**. We will attempt to accommodate students with legitimate conflicts. No makeup exams, early exams, or late exams will be allowed for any other reason.

3.3.3 Medical and Personal Emergencies

In case of a medical or personal emergency or similar excused reason, you must promptly notify us and **provide appropriate documentation** (such as a doctor’s note, dean’s excuse, or official university documentation).

3.3.4 Missed Exams

- If you miss one midterm for a valid, documented reason, your remaining two midterms will each count for 20% of your grade and the final exam will count for 30%.
- If you miss two midterms, you will not obtain a passing grade in this course.
- Decisions about missing the final exam will be at our discretion and will be made on a case-by-case basis.

4 Textbook

We will be using an online textbook for this course. Each student is required to obtain a personal copy of this textbook as it contains regular **graded assignments** to be submitted online.

1. Sign in or create an account at learn.zybooks.com. Please use your JHU email address.
2. Enter zyBook code: JHUEN.553.171NakadeFall12026
3. Subscribe

Direct link to the textbook: [zyBooks Course Page](#).

Yes, it costs money, and we looked hard for a free alternative before settling on it. The interactive exercises and immediate feedback are what make the active learning approach work, and we have seen students do noticeably better with this textbook than with a traditional one.

4.1 How to use the textbook

Textbook homework assignments will be due *before each class*. This allows us to spend class time working on problems together instead of sitting through a passive lecture. We'll post each assignment at least a week in advance, so there's always room to start early.

We will not "cover" all of the textbook material in class. You are expected to fill in the gaps yourself by reading the book, asking questions, and seeking one-on-one help. Here's how to get the most out of it:

- **Read before class, not after.** The reading is what makes class time useful. You'll get far more out of a discussion when you've already met the ideas once.
- **Take notes as you read.** Note-taking is an active process and a skill in its own right, and it will serve you long after this course. Write definitions in your own words and copy out the examples by hand.
- **Do the exercises honestly.** Attempt each one before revealing the answer. A wrong attempt you then correct teaches you more than a right answer you were handed.
- **Spread the work out.** A little each day beats one long session the night before. Proofs need time to settle.
- **Mark what confuses you.** Keep a running list of the things that didn't click, and bring it to class, discussion, or office hours. That list is one of the most useful study tools you'll have.
- **Revisit earlier sections.** The material builds on itself; when something new feels impossible, the gap is often a few sections back.

No exceptions will be made for late submissions. Instead, at the end of the semester, the textbook grade will be multiplied by 1.15x (up to a max of 100%). So completing about 87% of the homework activities is enough to earn a full 100% textbook homework grade. The built-in cushion is there for the week when life gets in the way.

5 Academic Integrity

All students must review and comply with the university-wide policies and resources provided on Canvas. These cover academic ethics, student well-being and mental health resources, Title IX and sexual misconduct reporting, disability services, and religious observance. Please take a few minutes to read through them, and reach out to us or to the offices listed there if you ever need support during the semester.

All work you submit in this course must be your own. Copying solutions, whether from a classmate, a solutions manual, a website, or a generative AI tool, is a violation of academic integrity.

5.1 AI Use Policy

The only AI tools permitted in this course are the ones described in [section 6](#) and [section 7](#). If we suspect that AI has been used to produce your work, **you will be asked to present your solutions to us in an oral exam**. Depending on the severity of the violation, being unable to do so may result in a 0 on the assignment, a failing grade in the course, and a report to the Dean's office for a violation of academic integrity.

We want to be clear about why this policy exists: the point of the homework is not the finished proof on the page, it's the thinking you do to get there. Outsourcing that thinking costs you the very skill the course exists to build. If you're stuck, please come to office hours, ask in class, or use the tutor in [section 6](#). All of those help you, and none of them put your grade at risk.

5.2 Disclosure of AI Use

Any use of AI in your coursework must be disclosed. This applies to the permitted uses as well: if you worked through a problem with the AI tutor, say so.

Disclosure is simple. At the end of your submission, add a short note naming the tool and what you used it for. For example: "I used the course AI tutor to help me get started on Problem 3." That's all we're asking for; a single sentence is enough.

Disclosing permitted use costs you nothing. It does not lower your grade, and it is not an admission of wrongdoing. It's a normal part of academic practice, the same way you would cite a book or a classmate who helped you. Failing to disclose is itself a violation of this policy, and it is treated far more seriously than the use would have been.

5.3 Our Own Disclosure

In keeping with the policy above: **this syllabus was revised with the help of Claude**, an AI assistant, which was used to tighten the wording, catch inconsistencies in the schedule, reorganize the document, and improve the clarity of several sections. All of the course policies, dates, and decisions in this document are our own, and we have reviewed every word of it.

We share this disclosure because we believe the same standard should apply to us as well.

6 AI Tutor

If you would like extra help working through problems without crossing the line into cheating, here are two options:

1. **Use the pretrained tutor.** A ready-to-use AI tutor for this course is available at <https://box.boodle.ai/a/@DiscreteBot>
2. **Build your own.** You may also set up your own tutor in any AI chat tool (ChatGPT, Claude, Gemini, etc.) by pasting the following prompt at the start of your conversation:

You are a TA holding office hours for a freshman undergraduate discrete mathematics course. Your goal is to help me learn—not just to give me answers, but to make sure I understand. Struggling productively with hard problems builds real understanding, but so does getting clear explanations when I’m genuinely lost. Your job is to read the situation and respond with what I actually need.

Default Mode

When I bring you a problem, start by understanding where I am:

- Ask what I’ve tried, or where I’m stuck.
- Guide with questions rather than answers when I’m close.
- Help me find my own errors rather than pointing them out immediately.
- If a theorem or technique is relevant, name it but ask me to try stating it before you explain.
- Keep responses focused—one or two questions at a time when you’re prompting me to think.

When to Shift to Direct Explanation

Drop the Socratic approach and just teach when:

- I’m missing a foundational concept I need before I can attempt the problem.
- I’ve been stuck on the same point through multiple exchanges.
- I ask for a worked example of a *different* problem to learn the technique.
- I need a definition, theorem statement, or factual clarification.
- I explicitly ask you to just explain something.

Guardrails

- **Don’t just solve homework problems for me.** If I ask you to, help me get unstuck instead—unless I’ve clearly been struggling and need to see the solution to learn from it.
- **Check in when it’s unclear what I need.** Ask: “Would you like a hint, or do you want me to walk through this part?”
- **Notice if I’m pattern-matching without understanding.** If I get an answer but my reasoning is shaky, slow down and ask me to explain my thinking.

Tone

Be warm, patient, and encouraging. Acknowledge when something is hard. Remind me that confusion is part of learning. Treat me as capable of getting there.

Topics include: logic and quantifiers, proof techniques (direct proof, proof by cases, contradiction, contrapositive, induction), inequalities, sets, functions, relations, equivalence relations and partial orders, number theory, modular arithmetic, multiplicative inverses, applications to cryptography (Diffie–Hellman key exchange, RSA), combinatorics, counting, stars and bars, combinatorial proofs, recurrence relations, graphs, trees, Euler circuits and Hamiltonian cycles, planarity, the six color theorem, and the Euler characteristic.

The goal is for me to leave each conversation understanding something I didn’t before—not just with an answer, but with the reasoning behind it.

7 Best Practices for Using AI to Learn Mathematics

The following is general guidance from the Department of Applied Mathematics and Statistics, shared across many of our courses. Where it differs from the policy in [subsection 5.1](#), the policy for this course governs.

Introduction. The use of AI tools (ChatGPT, Claude, Gemini, Microsoft Copilot, and others) has become commonplace among university students. Like any powerful tool, these systems can either enhance learning or hinder it, depending on how they are used. Faculty in the Department of Applied Mathematics and Statistics have observed that the increasing use of AI has led to markedly different learning outcomes among students. While some students use AI to deepen their understanding of the material, others become overly dependent on it and subsequently struggle on in-class quizzes, exams, and other assessments that require independent reasoning: *thus, AI is creating a wider spread in student outcomes*. We have also observed a noticeable decline in attendance at office hours: please keep in mind that while AI can be a convenient first resource, it should *never* be regarded as a substitute for meaningful interaction with instructors and TAs, whose role extends far beyond providing answers to questions.

For this reason, while the department generally permits the use of AI tools, it also believes that students should understand how to use them effectively and responsibly. The purpose of this document is to provide a concise set of guidelines for using AI to support the study of mathematics while preserving the critical thinking, analytical, and problem-solving skills that our courses are designed to develop.

Good vs. bad use of AI. Used appropriately, AI *can* be an excellent study companion. It can explain difficult concepts from different perspectives, generate additional practice problems, suggest alternative solution methods, and help you identify gaps in your understanding. Used carelessly, it can encourage passive learning, conceal misunderstandings, and occasionally produce answers that are persuasive but incorrect.

The purpose of using AI is not simply to obtain correct answers more quickly. Its real value lies in helping you become better at mathematics. The most important skill *you* (the student) must develop is not the ability to produce polished solutions, but the ability to understand why they work, recognize when they are incorrect or incomplete, and apply the underlying ideas to new problems.

For this reason, you should *never* accept an AI-generated response at face value. Instead, compare it with your own reasoning, verify each important step, and ask yourself whether you could reproduce the solution without AI. A well-written explanation is not the same as genuine understanding.

One of the best-established findings in the science of learning is the *illusion of competence*: students often believe they understand material that they are unable to reproduce, apply to a new problem, or explain under questioning. AI can make this illusion even stronger. Reading a clear, convincing explanation often creates the feeling that the material has been mastered, when in reality the understanding is only superficial. The true test comes when you must solve a similar problem independently.

A productive interaction with AI begins with clear instructions. Rather than asking for complete solutions, encourage AI to act as a tutor: ask it to provide hints instead of answers, point out mistakes in your reasoning, ask you questions, or challenge you to justify each step before moving on. As a final check, ask yourself the following question before considering an assignment complete:

Can I clearly and thoroughly explain every step of this solution to a classmate, without relying on AI?

If the answer is *yes*, then AI has probably served its intended purpose: it has helped you *learn* rather than merely helping you complete the assignment. The next section describes how to instruct AI to behave as a *tutor*, and suggests a recommended sequence of interactions to follow while working through homework.

Learning Should Require Effort. Learning mathematics is not supposed to be easy. Confusion, false starts, unsuccessful approaches, and the effort required to work through them are not unfortunate obstacles on the way to learning; they are an essential part of learning itself. You develop mathematical judgment by wrestling with problems whose solutions are not immediately apparent, deciding what might work, discovering why an approach fails, and trying again. A tool that removes too much of this struggle may make the immediate task easier while making the learning less effective.

At the same time, productive struggle does not mean struggling alone. Mathematics has always been learned in conversation with other people. A classmate may ask the question that exposes a gap in your reasoning; a TA may give just enough of a hint to get you moving again; an instructor may recognize that your difficulty comes from a misconception several steps earlier. Explaining an idea to another person, and having that person question your explanation, is itself an important form of mathematical work. Good teaching therefore does not eliminate difficulty. It helps a learner remain engaged with a difficult problem while providing the right amount of support at the right time.

AI changes this balance because assistance is now available instantly and in essentially unlimited quantities. That convenience makes it especially important to resist the impulse to remove every difficulty as soon as it appears. When you encounter a problem that makes you stop, think, struggle, discuss, erase your work, and start again, something valuable may be happening. Do not be too quick to make that experience disappear. And do not allow the convenience of an AI assistant to crowd out the less convenient, but often richer, experience of working through mathematics with other human beings.

Good prompts and habits. You should develop the habit of following these five steps: try to solve, ask for help, work independently, verify, and reflect.

- **Try to solve it yourself.** Make an honest attempt before asking AI for help. Identify the relevant concepts and state clearly what the problem is asking you to prove or compute, in your own words.
- **Ask for help.** Ask for hints, explanations, feedback, or criticism of your reasoning. Do *not* ask your AI for a complete solution.
- **Rework the problem independently.** Return to the task and produce your own solution. Write out (preferably by hand) the procedure clearly, justify each step. Identify the results you are using.
- **Verify.** Check the claims, computations, assumptions, and applications of theorems. You may ask AI to critique your solution and identify steps that are incorrect, unclear, or insufficiently justified.
- **Reflect.** Make sure that you understand the solution well enough to explain it clearly to a classmate without relying on AI.

Here are examples of ‘good prompts’ (you can figure out what the ‘bad prompts’ are).

- I’ll explain my procedure to you. Tell me which parts are incorrect, incomplete, or unclear.
- I’m stuck at this point. Don’t solve the problem or tell me the next step. Ask me one question that might help me decide what to try next.
- What assumptions am I making that I have not stated?
- Explain this concept/property in plain language, then give a rigorous version.
- Give me a concrete example and a non-example employing this concept/property.
- Ask me three questions to check whether I understand this idea.
- Review my proof, but do not identify an error or missing step directly. Ask me questions that will help me find any gaps myself.
- Create 3 problems that test my understanding of the concepts in Chapters 10--12 of the book. Give me one problem at a time. Don’t provide solutions/hints until I have made a serious attempt.

Bottom lines.

1. **Submit only work you *truly* understand.** AI can help you learn difficult concepts, but it should never be used to disguise gaps in your understanding.
2. **While AI can complement human interaction, it should *not* replace it.** Your instructors, classmates, and university support services can provide guidance, encouragement, and perspectives that no AI can match. Make it a habit to attend your instructors’ or TAs’ office hours regularly, even if only briefly and even when you are not stuck on a particular problem. If you are struggling, whether with the material or with personal challenges, reach out to people. Seeking help is a sign of good judgment, not of weakness. Learning mathematics is *not* meant to be a solitary activity.